



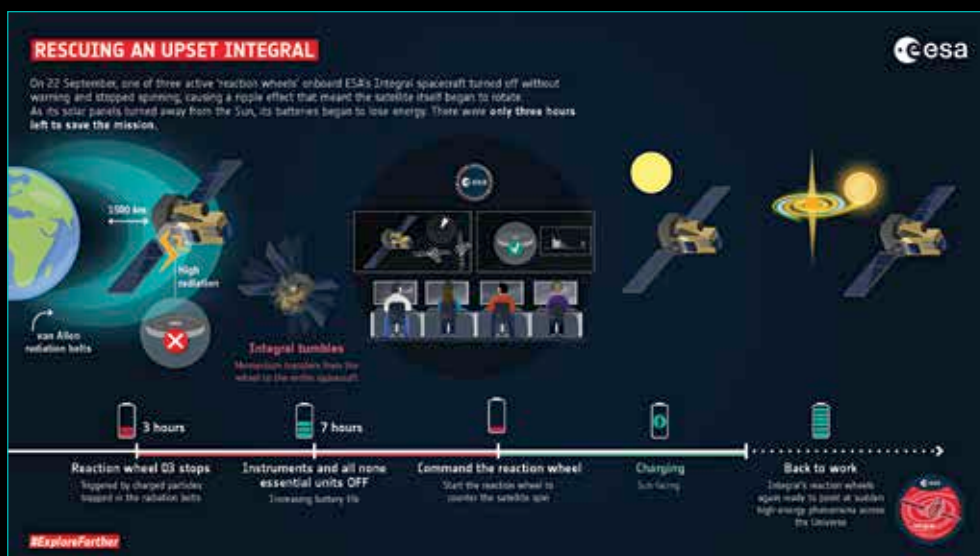
Fastest quantum supercomputer

Chinese scientists produced a super computer 10mn times faster than current record holder. <https://digit.in/nov21-13>



Unknown disease kills reefs

A mysterious disease is killing Florida's coral reefs. <https://digit.in/nov21-14>



Three hours to save integral

direction to make observations. This required planning and scheduling the science targets for the observatory, to make sure that what the satellite could point towards was closely tracked to how the satellite stored momentum and juggled it between the reaction wheels. By September 2020, the satellite was back to conducting science, and since then the team continues to improve the z-flip maneuver, coming up with more innovative and efficient ways of juggling the energy between the reaction wheels, and giving the satellite more operational flexibility at the same time. There are two important things to note here, this satellite was brought back from the dead during the middle of a global pandemic and lockdowns across Europe. The ESA managed to rescue the satellite while working remotely. The other thing is that the complicated z-flip maneuvers were first simulated on computers, before being relayed to the satellite. The version used for the actual maneuver was different from the one used in the simulation, and the sequence of movements have been constantly improved since then. For the ESA engineers, this went to show that they do not need to be absolutely accurate at all times, and being approximate is good

enough in some cases. As gamma ray events tend to be short lived, it is very important for the satellite to be nimble enough to turn to a portion of the sky on demand. This flexibility has been reduced since the propulsion system ran out, and the satellite now operates in the new "slew" mode, which is the process of going from pointing in one direction, to pointing in another.

At some time in the distant past, a high energy particle from some unknown part of the sky was trapped in one of the two Van Allen belts of the Earth, donut shaped regions around the planet where high energy particles keep circulating. One of these charged particles struck a sensitive part of Integral's sensitive electrical components on 22 September 2021. This caused what is known as a Single Event Upset (SEU), causing a rare state change that disrupted the performance of the satellite. One of the reaction wheels just stopped turning, dumping all the collected energy into the satellite. Because of the law of conservation of energy, the only logical place for the energy to go was the satellite itself, which began to turn instead of the reaction wheels. This triggered the satellite to revert to its programmed

emergency safe mode, which was no longer reliable without the propulsion system. Ground control managed to reactivate the malfunctioning wheel, but the satellite was turning once every 21 minutes, rotating five times the maximum rate under normal control. While spinning, the satellite was also wobbling in unpredictable ways.

The team had three hours to save the spacecraft before its batteries ran out,

making it a derelict. At the same time, the data coming down from the spinning spacecraft was in short periods, with a small window to transmit the necessary sequence of maneuvers to return the spacecraft to normal operations. Various instruments and non critical components were used to conserve energy, and increase the available time to six hours. The state of the reaction wheels was analysed, and a sequence of commands were relayed to the satellite, to change the speed and put a brake on the spinning. Over a period of three hours, these commands were executed to bring the satellite back under control, and charge up the batteries. At this point, for some unknown reason, the star tracker failed and the satellite started spinning out of control again. The ground control team accounted for the star tracker and the same series of commands was used to again bring the satellite under control. The entire recovery operation was conducted with most of the personnel working from home. Integral is now back in action, and is even participating in multimesenger astronomy with the gravitational wave observatories, a science that did not exist when the satellite was first deployed. **d**